

Newborn Sepsis with Apneas



Section I: Scenario Demographics

Scenario Title:	Newborn Sepsis with Apneas
Date of Development:	29/AUG/2016
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☒ Seniors (PGY ≥ 3) ☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Rob Woods
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To review the management of the septic newborn
CRM Objectives:	1) Effectively lead a newborn resuscitation and seek input from others 2) Use effective communication strategies, including closed loop communication and summarizing events for team
Medical Objectives:	1) Recognize sepsis and shock in the newborn 2) Rapidly estimate pediatric weight for medication dosing 3) Obtain efficient vascular access and initiate fluid resuscitation 4) Initiate early broad-spectrum antibiotics. 5) Recognize recurrent apneas as an indication for intubation and plan for a safe intubation accordingly

Case Summary: Brief Summary of Case Progression and Major Events

A 3-day-old term male infant is brought to the ED by EMS after being seen at their Family Physician's office with a low temperature (33.1°C). The child has been feeding poorly for about 12 hours, and has vomited twice. He is lethargic on examination and poorly perfused with intermittent apneas lasting ~ 20 seconds. He requires immediate fluid resuscitation and broad-spectrum antibiotics. His perfusion will improve after IVF boluses, however the apneas will persist and necessitate intubation.

References

<http://emedicine.medscape.com/article/972559-overview>, PALS & APLS courses



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☐ Computerized Mannequin	Realism:	☒ Conceptual
	☒ Neonatal Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates			
Brief Description of Role			
Resus Nurse	Hooks patient up to monitors, provides additional vitals if requested (Blood sugar 2.7, Temperature 33.1°C). Frequently describes the patient as looking pale/mottled, rapid HR, prolonged capillary refill.		
B. Required Monitors			
☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☒ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☐ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☒ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☒ Other: Broselow tape	
☒ Intraosseous Set-up	☒ LMA	☒ Other: Neonatal warmer	
D. Moulage			
Infant mannequin to be described or dressed as a 3-day old male infant, weighing 4kg at birth.			
E. Approximate Timing			
Set-Up:	10 min	Scenario:	15-20 min
		Debriefing:	20-30 min

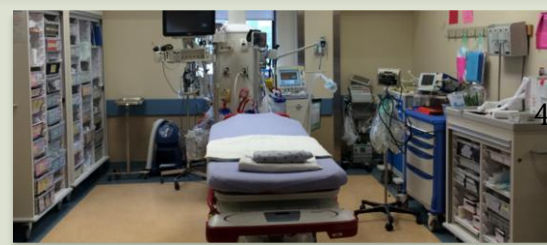
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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case			
To be stated by the Paramedic with the Resus Nurse at bedside: "We picked up this term 3-day old male infant at their GPs office. Mom reports poor feeding for the past 12 hours, and two episodes of vomiting. They took him to the GPs office this morning and they found the temperature to be quite low at 33.1°C. They called us concerned about sepsis. We were only 5 minutes away so we have not obtained IV access. We did obtain a glucose level of 2.7. The child is lethargic and has very poor perfusion – peripheral cap refill is 7 seconds. We don't have a cuff to get an accurate BP but the HR is 190."			
B. Patient Profile and History			
Patient Name: Jake Brown		Age: 3 days	Weight: Birth weight 4kg Grey on Broselow tape
Gender: ☒ M ☐ F		Code Status: Full	
Chief Complaint: Low temperature at GP.			
History of Presenting Illness: Poor feeding x12 hours, vomiting.			
Past Medical History:	Term infant. Normal pregnancy. GBS+ Premature ROM. Got Abx before delivery.	Medications:	None
Allergies: None			
Social History: Lives with Mom and Dad			
Family History: None			
Review of Systems:	CNS:	Nil.	
	HEENT:	Nil.	
	CVS:	Nil.	
	RESP:	Nil.	
	GI:	Poor feeds x12 hours. Vomited twice in 24 hours.	
	GU:	Only 2 wet diapers today.	
	MSK:	Nil.	INT: Nil
C. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☐ Monitor On, no data displayed	☒ Monitor on Standard Display
HR: 190/min	BP: 50/palp	RR: 40/min with apneas	O ₂ SAT: 93%, drops to low 80s after apneas
Rhythm: NSR	T: 33.1°C	Glucose: 2.7 mmol/L	AVPU: Pain only
General Status: Eyes closed, cries weakly with external stimulation			
CNS:	PERLA, 3mm bilaterally.		
HEENT:	Normal		
CVS:	Cap refill 7 seconds peripherally, 5 seconds centrally, mottled extremities		
RESP:	Shallow resps with 20-30 second apneas every 5 mins		
ABDO:	Soft, not distended. No rebound or guarding.		
GU:	Nil.		
MSK:	Nil.	SKIN:	Mottled extremities

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Section VI: Case Progression

Scenario States, Modifiers and Triggers

Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus HR: 190/min BP: 50/palp RR: 40/min, shallow O ₂ SAT: 93% RA T: 33.1°C rectal	GCS - AVPU	<u>Learner Actions</u> ☐ Primary survey and coordination of team ☐ Monitors ☐ Establish iv or IO access ☐ Determine weight of child (birth weight, Broselow tape, age based) ☐ Push dose fluid boluses (20ml/kg x3) ☐ Re-check glucose (1.8mmol/L) ☐ Correct hypoglycemia (5cc/kg of D10W) ☐ Broad-spectrum ABX (cefotaxime 100mg/kg + ampicillin 100mg/kg)	<u>Modifiers</u> - 1 st fluid bolus → CRT 5 sec peripheral, 3 sec central, HR 175, BP 60/30 <u>Triggers</u> - 2 nd bolus → 2. Apneas - No glucose given by 5 min → 3. Seizure
2. Apneas HR: 175/min BP: 65/35 RR: Apnea x20 sec then back to 40 O ₂ SAT: drops to 81% with apnea then back to 93%	GCS - AVPU	<u>Learner Actions</u> ☐ Recognize apnea with desaturation, provide supplemental oxygen ☐ Administer 3 rd fluid bolus ☐ Establish IV access ☐ Draw labs including VBG, lactate, cultures ☐ Re-check glucose (9.7mmol/L) ☐ Portable CXR ☐ Call for high flow O ₂ ☐ Prepare for definitive airway management (including push-dose pressor at bedside)	<u>Modifiers</u> - No recognition of apnea, no O ₂ applied, or no preparation to intubate → Repeat apneic episode <u>Triggers</u> - 3 rd bolus given and O ₂ applied OR 5 min into state → 4. Persistent Apnea
3. Seizure HR: 180 BP: 85/45 O ₂ SAT: 88%	Patient actively seizing	<u>Learner Actions</u> ☐ Re-check glucose (1.8mmol/L) ☐ Correct hypoglycemia (5cc/kg of D10W)	<u>Modifiers</u> - No glucose check → RN cues <u>Triggers</u> - Glucose replaced → 2. Apneas
4. Persistent Apnea HR: 160 BP: 70/45 RR: apnea O ₂ SAT: drops to 81% then slowly to 75%	GCS - AVPU	<u>Learner Actions</u> ☐ Assist ventilation with BVM until ready to intubate ☐ Administer appropriate induction agent ± paralytic + atropine ☐ Intubate using appropriate tube size (3 cuffed or 3.5 uncuffed) and depth (9cm)	<u>Modifiers</u> - BVM prior to intubation → O ₂ SAT 99% <u>Triggers</u> - No BVM or intubation delayed >3 min → 5. PEA - Successful intubation → 6. Resolution
5. PEA Rhythm: sinus HR: 45/min BP: 0 RR: 0	Pulseless	☐ Initiates CPR with appropriate 15:2 ratio ☐ ± Crash intubation ☐ Epi iv at 0.01mg/kg	<u>Triggers</u> - Successful intubation → 6. Resolution - 3 rounds CPR no intubation → 4. Persistent Apnea
6. Resolution HR: 150 BP: 70/45 O ₂ SAT: 98% RR: 40	GCS - AVPU	<u>Learner Actions</u> ☐ Consult Peds ICU ☐ Post-intubation CXR ☐ Cath urine for urine culture	End Case PRN

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results

No blood work is provided during the case.

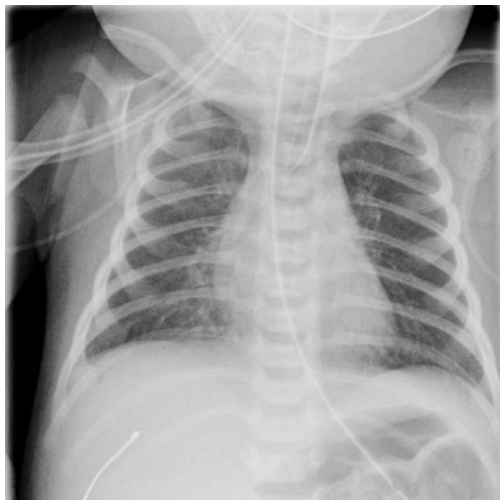
Images (ECGs, CXRs, etc.)

Initial CXR



(CXR source: <http://emedicine.medscape.com/article/414608-overview>)

Post-Intubation CXR



(CXR source: <https://radiopaedia.org/articles/neonatal-pneumonia>)

Ultrasound Video Files (if applicable)

None required.

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) Did you feel that your team leader had control over the room? Did he or she ask for input? 2) Do you feel you communicated well as a team? 3) What were your priorities in managing this patient? How did they change as the case progressed? 4) What was the cause of this patient's abnormal vitals signs and level of consciousness? Why was the patient having apneas? 5) How do you get vascular access in a newborn? 6) What infectious agents are you trying to treat with your antibiotic choices? Which is the more important antibiotic to give? 7) How do you manage hypothermia in the context of sepsis? What tactics did you use in this patient?	
Key Moments	
Recognition of toxic neonate and need to initiate immediate fluid boluses.	
Recognition of apneas and initial management steps (fluid boluses, O2, high-flow O2)	
Recognition of need to intubate	